

DBMS Lab Assignment: ER Diagram Design

Instructions:

1. Draw ER diagrams for the following problems using **draw.io**
2. Clearly identify entities, attributes, relationships, and cardinalities.
3. Use rectangles for entities, diamonds for relationships, and ovals for attributes.
4. Underline primary keys and mention relationship attributes wherever applicable.

Problem 1: Student–Course Registration System

A college maintains information about students and courses. Students can enroll in multiple courses, and each course can have many students. For each enrollment, the semester and grade should be recorded.

Problem 2: Library Management System

A library issues books to its members. A member can issue multiple books, but a book can be issued to only one member at a time. The issue date and return date must be stored.

Problem 3: Online Shopping System

An online shopping website maintains information about customers and their orders. Each customer can place multiple orders, but each order is placed by exactly one customer.

Problem 4: Hospital Appointment System

A hospital manages patients and doctors. Patients can book appointments with doctors. Each appointment has a specific date and time.

Problem 5: Employee–Department System

A company has multiple departments and employees. Each employee works in only one department, while a department can have many employees.

Problem 6: University Examination System

A university conducts examinations for different courses. Students enroll in courses and appear for exams. Each exam is conducted for a specific course on a particular date. Students receive marks and grades for each exam. A course may have multiple exams and students may appear in multiple exams.

Problem 7: Railway Reservation System

A railway reservation system maintains information about trains, passengers, and tickets. Each train runs between a source and a destination and has multiple coaches. Passengers can book tickets for trains. A ticket can include multiple passengers, and each passenger has a seat number. The system should also store the journey date and ticket booking status.